FEMS 2025
EUROMAT

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14 – 18 September 2025

High-throughput analysis of loop dynamics in irradiated MPEAs using deep learning

Camilo A. F. Salvador et al.

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Université Paris-Saclay, CEA Service de recherche en Corrosion et Comportement des Matériaux, (SRMP).

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HT analysis of loop dynamics in irradiated MPEAs using DL
(Background)

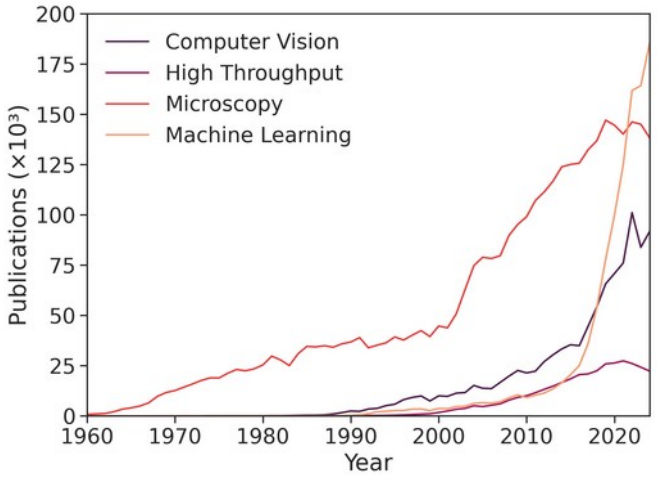


Fig. 1. Evolution of publications per year on relevant topics for this presentation; The data was obtained from the Web of Science in April 2025.

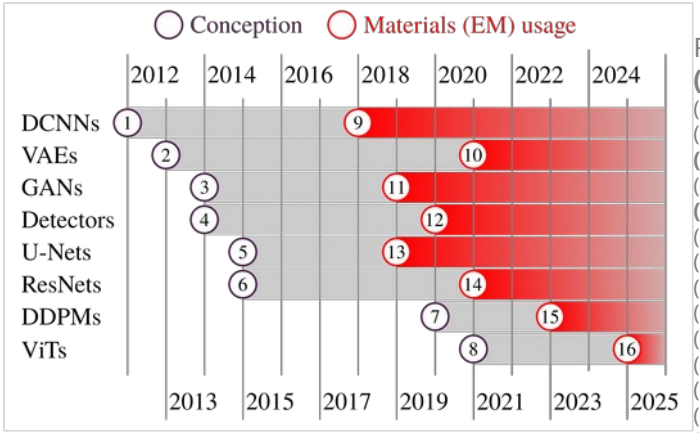


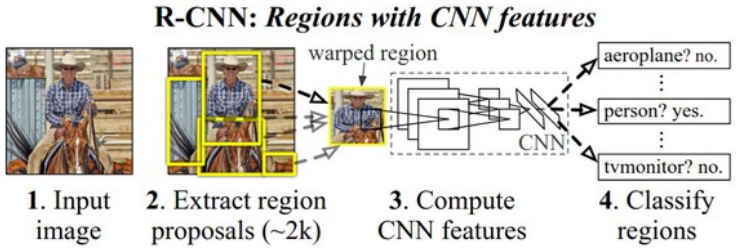
Fig. 2. Timeline of the most critical model families in computer vision and their respective adoption (in materials science electron microscopy).

References:

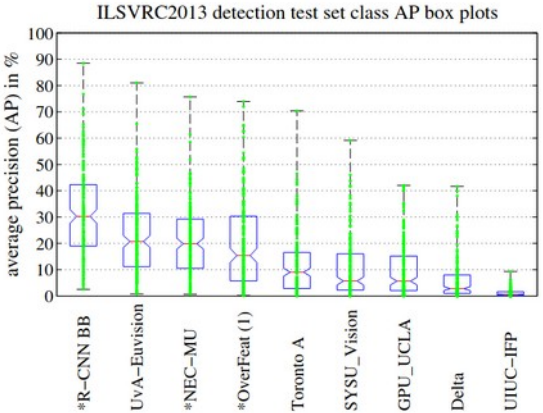
(1) Krizhevsky et al. (2012);
(2) Kingma & Welling (2013);
(3) Goodfellow et al. (2014);
(4) Girshick et al. (2014);
(5) Ronnenberger et al. (2015);
(6) He et al. (2016);
(7) Ho et al. (2020);
(8) Dosovitskly et al. (2021);
(9) Madsen et al. (2018);
(10) Zhong et al. (2021);
(11) de Haan et al. (2019);
(12) Okumev et al. (2020);
(13) Yao et al. (2019);
(14) Saxena et al. (2021);
(15) Azqadan et al. (2023);
(16) Whitman and Latypov (2025).

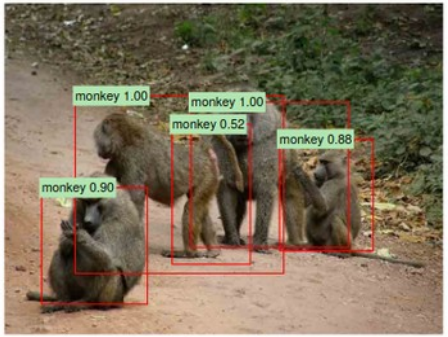
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HT analysis of loop dynamics in irradiated MPEAs using DL
(Detectors in Computer Vision)



Detectors in CV
R-CNN (2014)
Faster R-CNN (2015)
Mask R-CNN (2016)
(...)





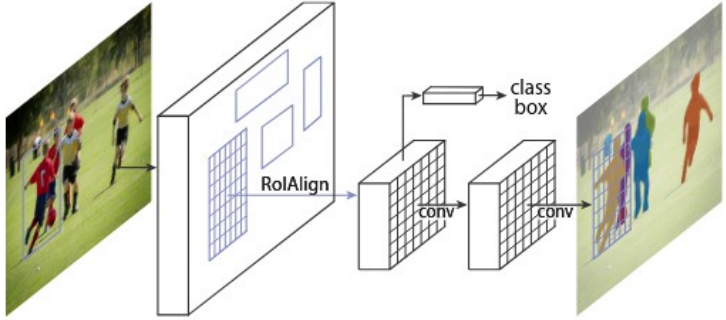




R-CNN
Girshick et al., CVPR 2014.
Images from the arXiv version;
arXiv:1311.2524v5 [cs.CV]

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HT analysis of loop dynamics in irradiated MPEAs using DL
(Instance-first methods with segmentation)



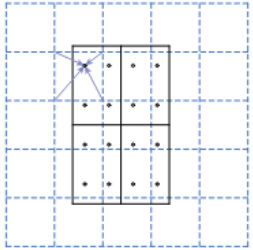


Figure 3. **RoIAlign:** The dashed grid represents a feature map, the solid lines an RoI (with 2×2 bins in this example), and the dots the 4 sampling points in each bin. RoIAlign computes the value of each sampling point by bilinear interpolation from the nearby grid points on the feature map. No quantization is performed on any coordinates involved in the RoI, its bins, or the sampling points.

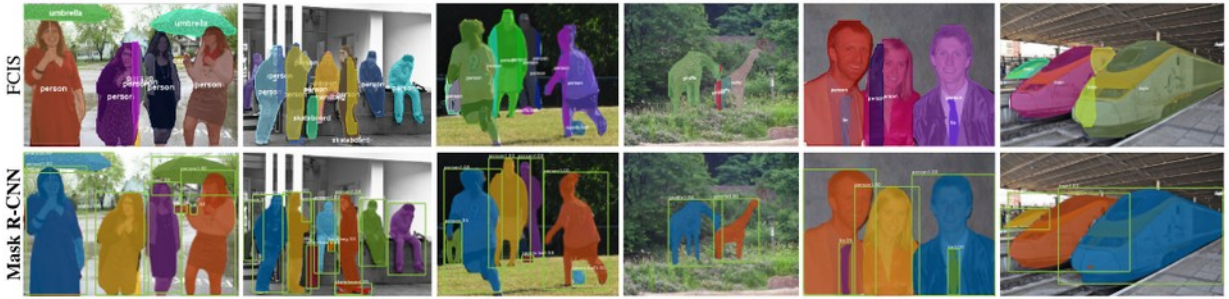
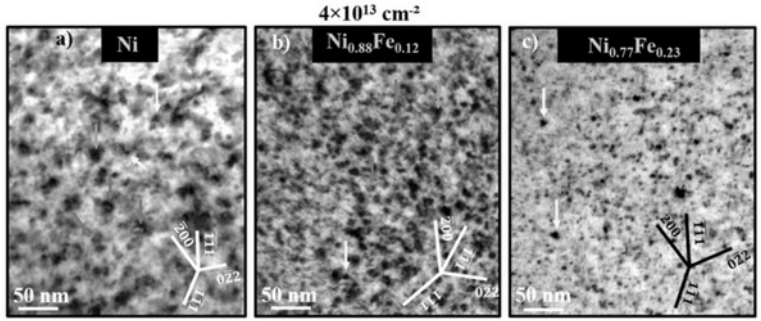


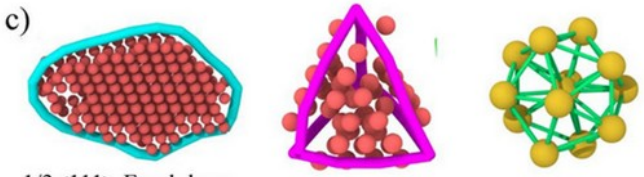
Figure 6. FCIS+++ [26] (top) *vs*. Mask R-CNN (bottom, ResNet-101-FPN). FCIS exhibits systematic artifacts on overlapping objects.

Mask R-CNN
He et al., CVPR 2017.
Images from the arXiv version
arXiv:1703.06870v3 [cs.CV]

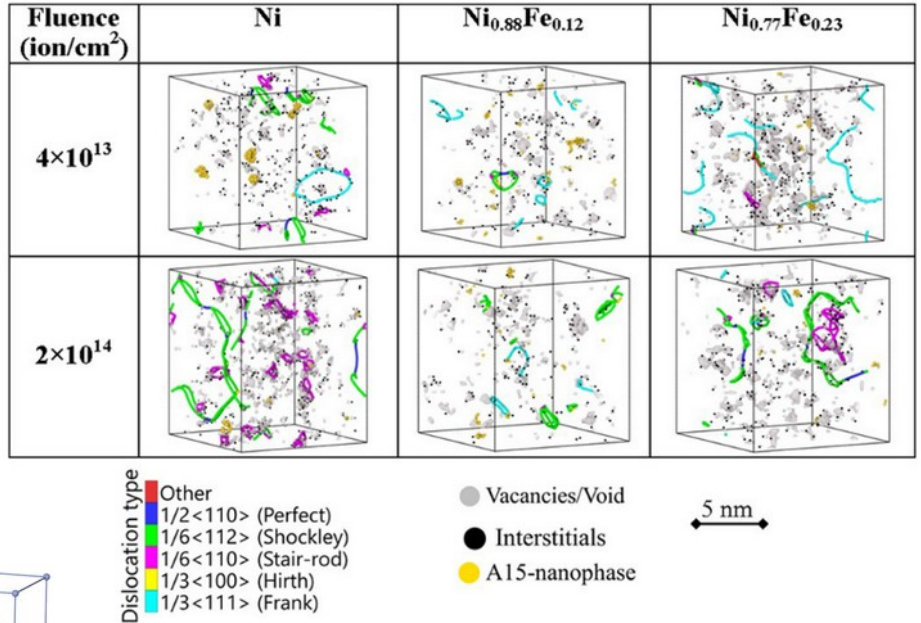
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HT analysis of loop dynamics in irradiated MPEAs using DL
(materials under irradiation)





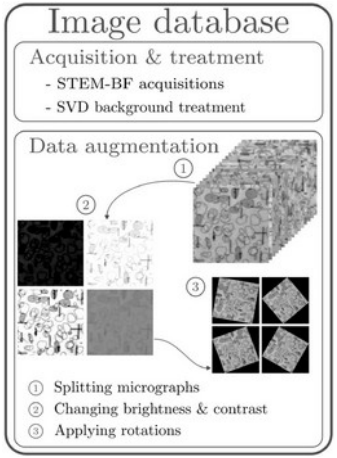


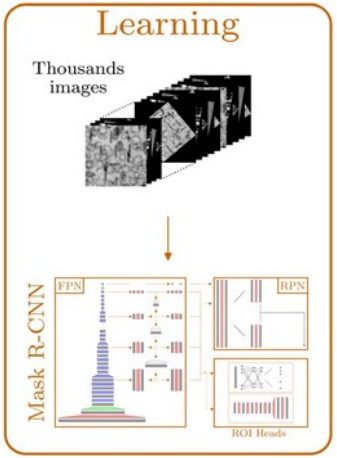


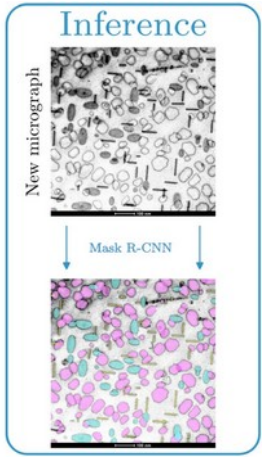
(a few images adapted from)
Wyszkowska et al., Nanoscale, 2025, 17, 15841

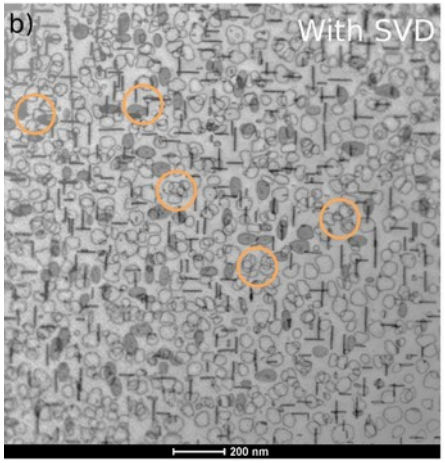
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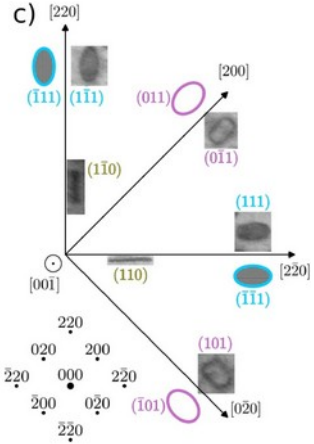
HT analysis of loop dynamics in irradiated MPEAs using DL
(preliminary works)









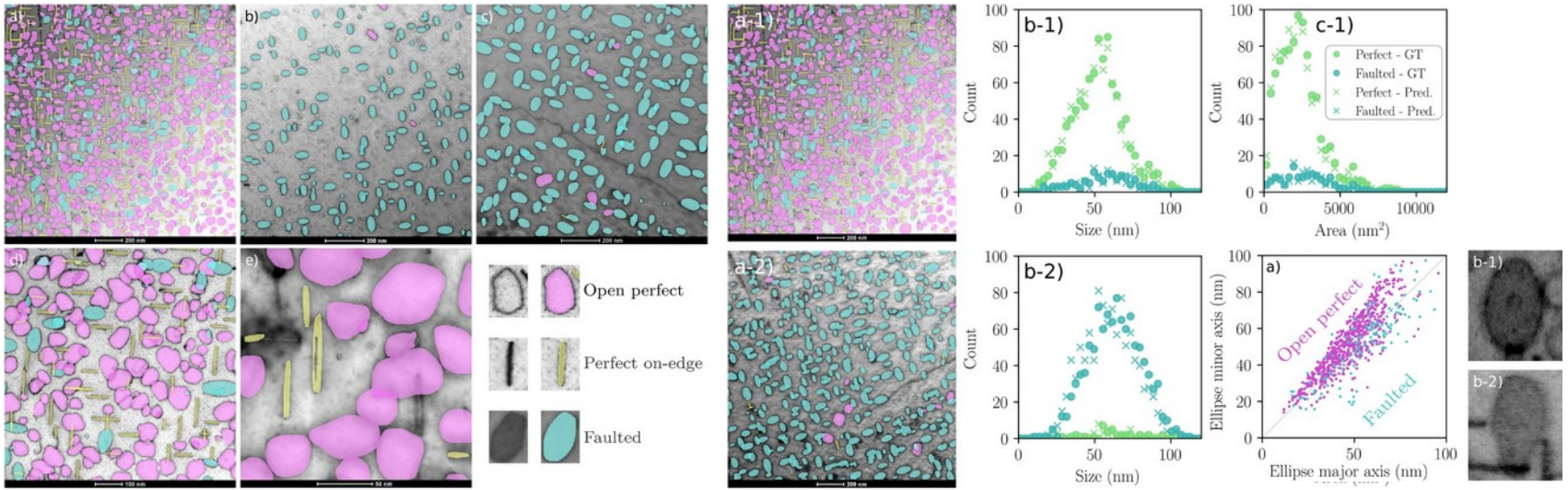


Bilyk et al., 2024
Sci. Rep. 14, 25168 (2024)

- 15Cr-46Fe-17Mn-22Ni (at.%) (Y3)
- 16Cr-37Fe-13Mn-34Ni (at.%) (ES1)
- Irradiation experiments with Fe^{+} and Ni^{2+} ions
- T = 550 °C

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HT analysis of loop dynamics in irradiated MPEAs using DL
(preliminary works)

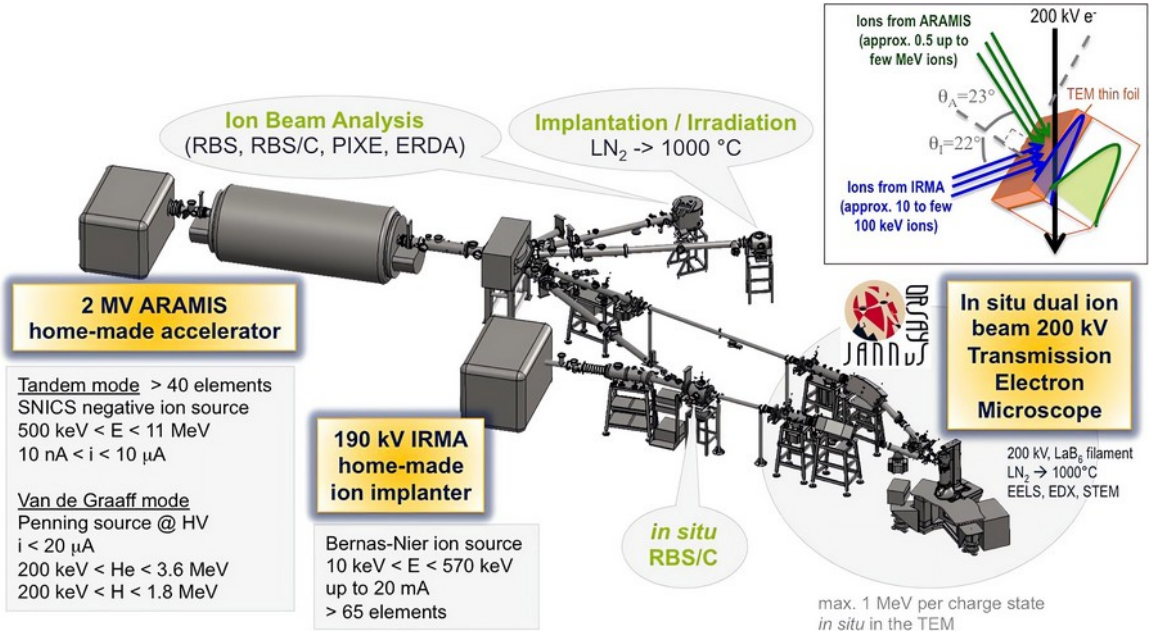


Bilyk et al., 2024
Sci. Rep. 14, 25168 (2024)

- 15Cr-46Fe-17Mn-22Ni (at.%) (Y3)
- Top (2 MeV Fe^{+}) versus bottom (6 MeV Ni^{2+} ions)
- 1.6 10^{-4} dpa/s up to 0.2 dpa
- Images are presented into a simplified form;

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HT analysis of loop dynamics in irradiated MPEAs using DL
(irradiation experiments)

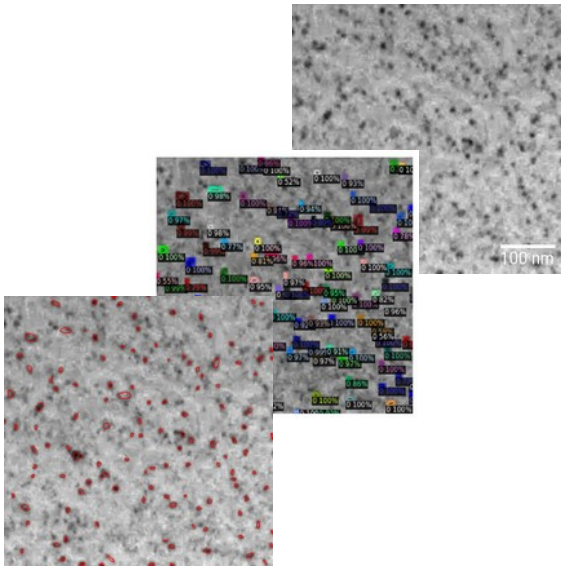


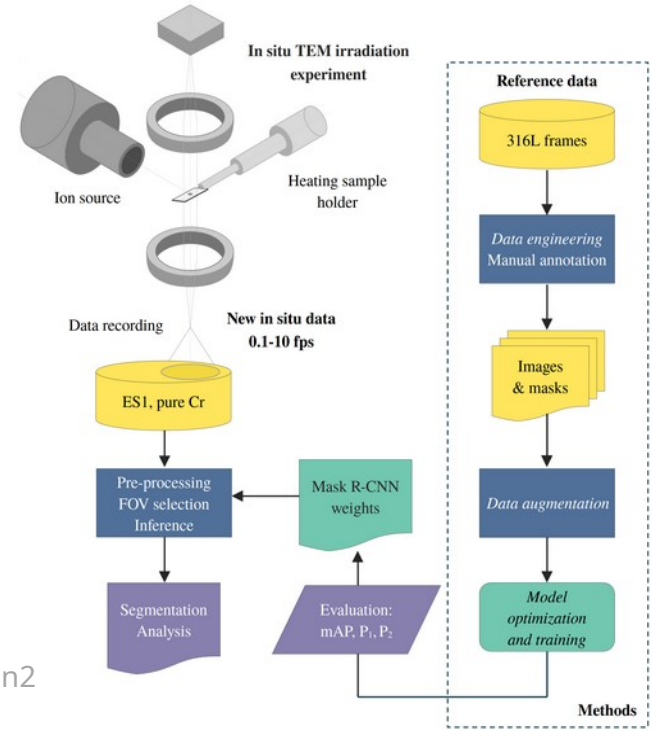
The JANNuS-Orsay facility
Gentils and Cabet, 2019
Nucl. Instrum. Methods Phys. Res., Sect. B, 447 (2019)

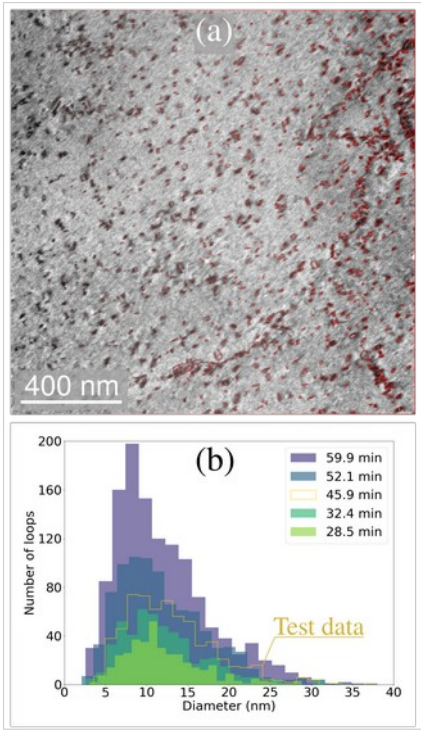
- Spatial resolution of 0.25 nm.
- 2048 x 2048 px CCD camera (up to 30 fps)
- Additional wide-area imaging camera

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Proposed framework
(https://github.com/ai-atoms/creme)





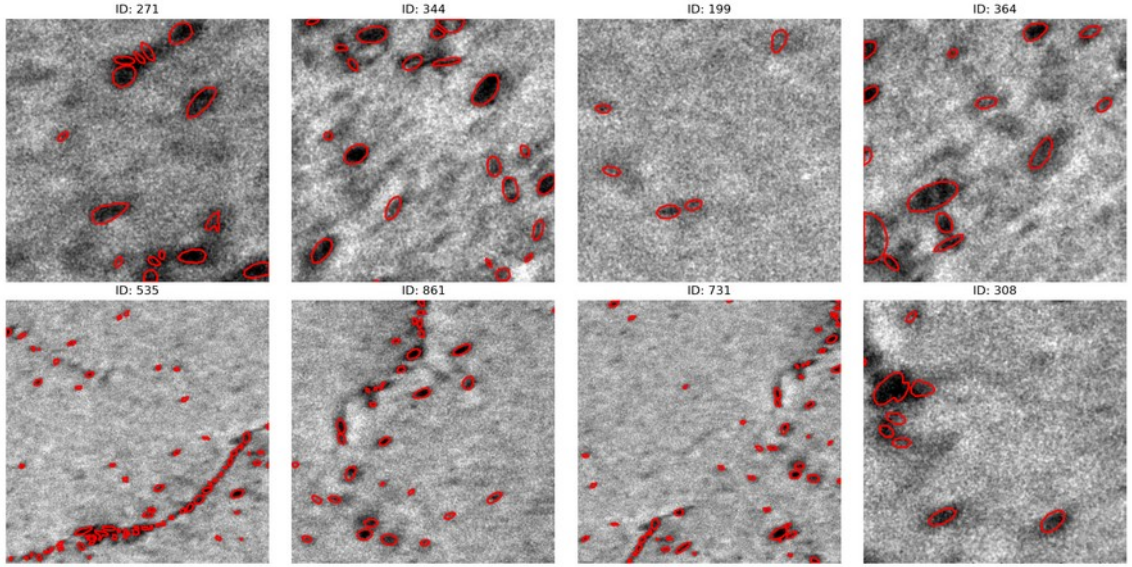


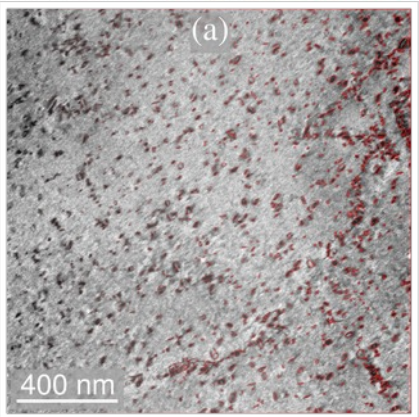
Detectron2 (Wu et al., 2019)
github.com/facebookresearch/detectron2

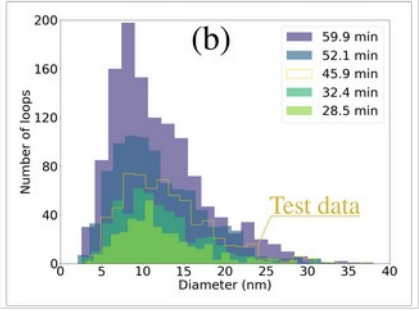
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Reference data
(Dataset: 43835 + 10975 loops)

316L, 550 °C, 2 MeV Ni^{2+} ions, ($2.8x10^{11}$ ions/cm^2.s)







FOV = 800 nm (2%), 533 nm (28%), 400 nm (31%), 200 nm (39%)

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Model training and evaluation

Backbone	Parameters (M)	FOV (nm)	Threshold	Run time (s)	Relative errors (%)				AP (%)	P_1	P_2
					F_{re}	N_{re}	D_{re}	$D_{m,re}$			
ResNeXt-101	89.46	200	0.8	11.0	1.01	-3.68	-10.03	-3.54	25.0	95.1	83.6
-	-	400	0.7	3.0	-8.10	-0.26	-22.81	-21.33	33.0	94.4	60.6
-	-	533	0.6	1.9	-9.88	2.59	-30.13	-30.57	36.6	88.0	49.8
Resnet-50	26.58	200	0.9	6.6	0.10	-0.15	-13.27	-6.01	19.6	97.2	81.4
-	-	400	0.8	1.9	0.94	-4.07	-15.71	-12.54	28.1	93.9	70.7
-	-	533	0.7	1.2	2.37	-1.77	-20.33	-19.20	33.0	92.9	63.2
Resnet-18	13.77	200	0.9	4.7	-19.79	-6.48	-13.71	-3.99	28.2	89.8	77.5
-	-	400	0.8	1.4	-16.62	-18.54	-14.32	-10.13	30.7	85.4	62.7
-	-	533	0.7	0.9	-13.02	-11.74	-21.09	-19.92	33.2	82.1	55.8

With:
F = foreground percentage
N = number of particles
S = particles' sizes
(ellipses with D and D_m)

$$Y_{\mathrm{re}} = \left(\frac{Y_{\mathrm{pred}} - Y_{\mathrm{gt}}}{Y_{\mathrm{gt}}} \right) \times 100 \qquad (1)$$

$$P_1 = 100 - 6 \left(\frac{1}{1+|F_{\mathrm{re}}|} + \frac{1}{1+|N_{\mathrm{re}}|} + \frac{4}{1+|S_{\mathrm{re}}|} \right)^{-1} \qquad (2)$$

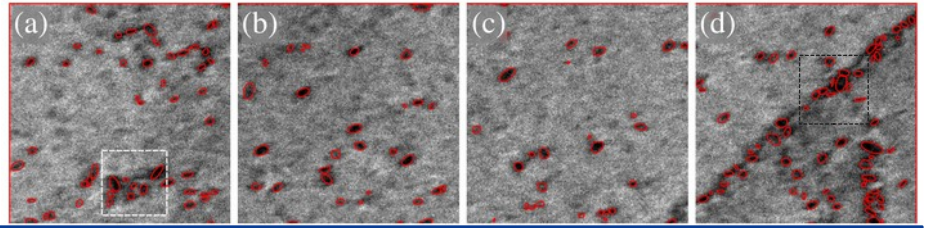
$$A_i = \pi \cdot D \cdot D_m \qquad (3)$$

$$P_2 = A_{\mathrm{tot}} = \langle N \rangle \cdot \pi \cdot \langle D \rangle \cdot \langle D_m \rangle \qquad (4)$$

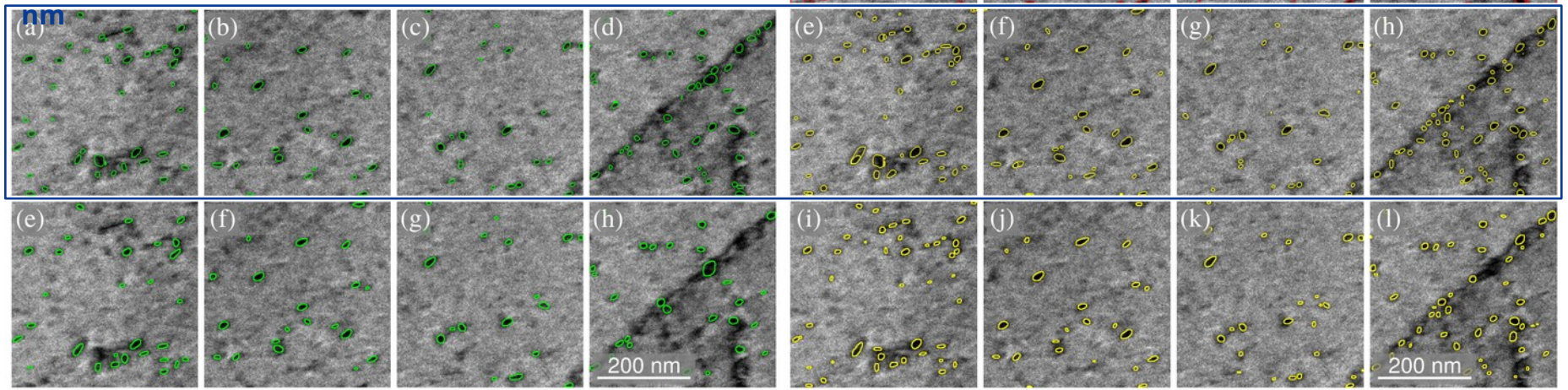
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Discussion, 316L validation

316L, 550 °C, 2 MeV Ni^{2+} ions
($2.8x10^{11}$ ions/cm^2.s)
Frame captured at 45.9 min



FOV = 200 nm



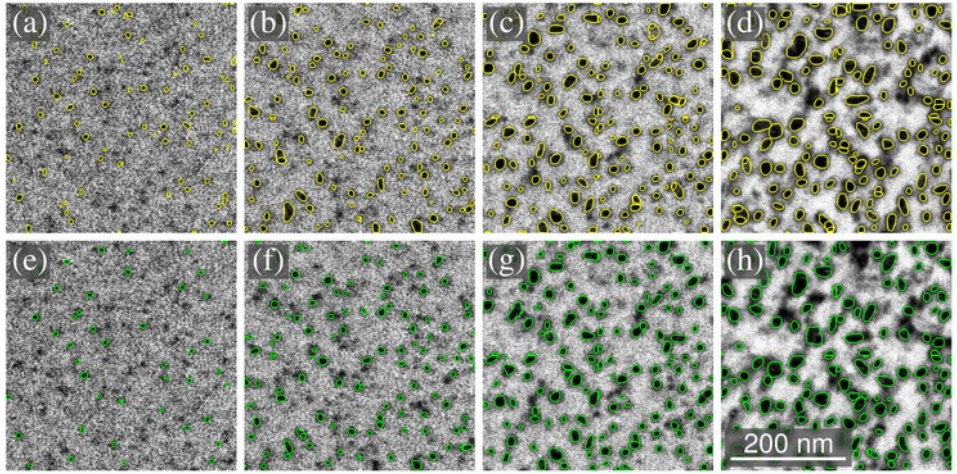
ResNet-50

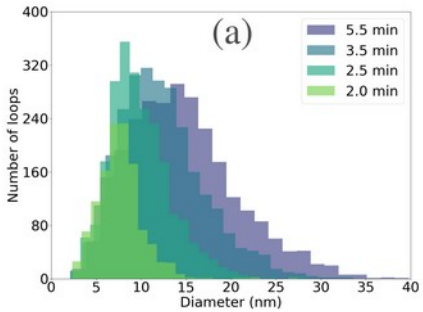
Ground-truth *versus* ResNeXt-101 (32x4d)

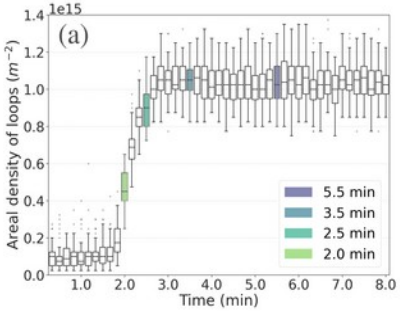
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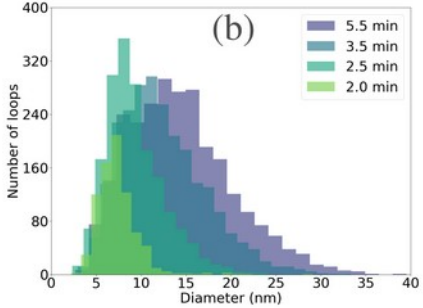
Use case 1, ES1 alloy
(rich datasets lead to robust models)

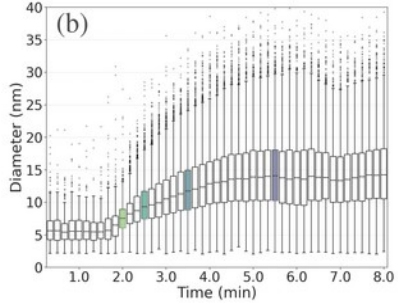
500 °C, 2 MeV Ni^{2+} ions
($4.2x10^{12}$ ions/cm^2.s)







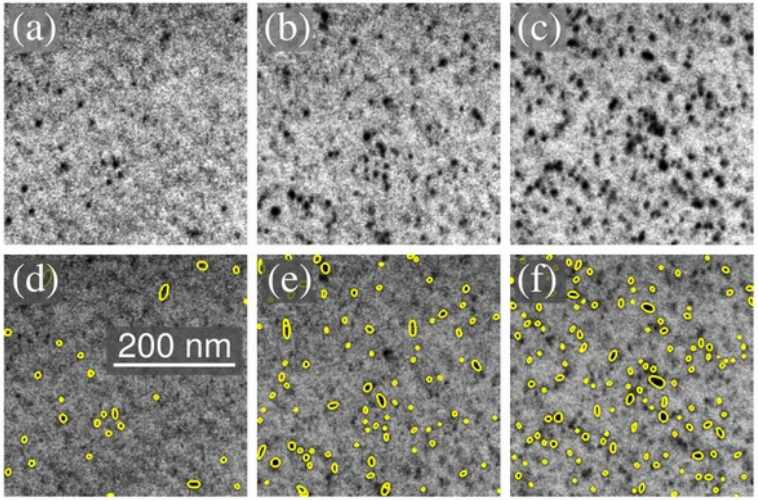


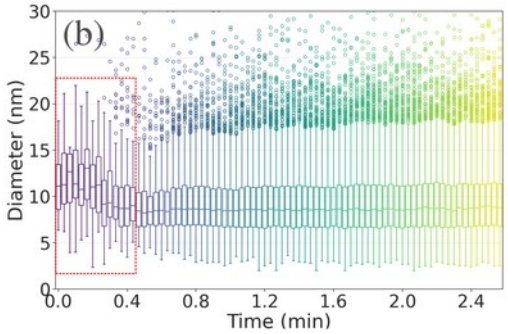


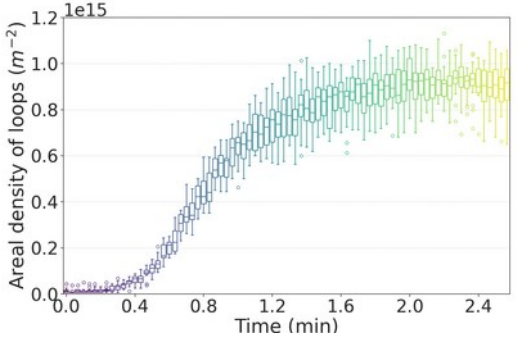
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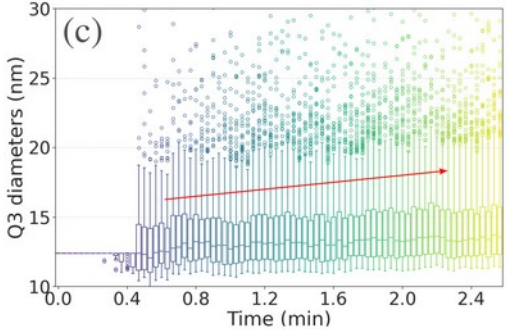
Use case 2, Pure Cr
(no need of background treatment)

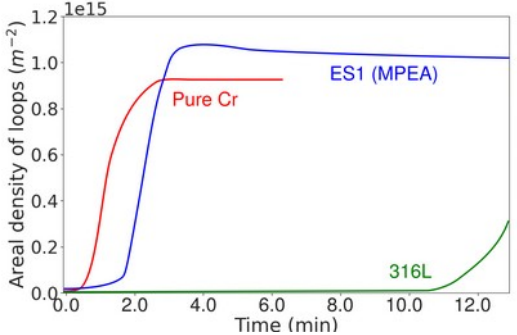
500 °C, 2 MeV Ni^{2+} ions
($4.2x10^{12}$ ions/cm^2.s)





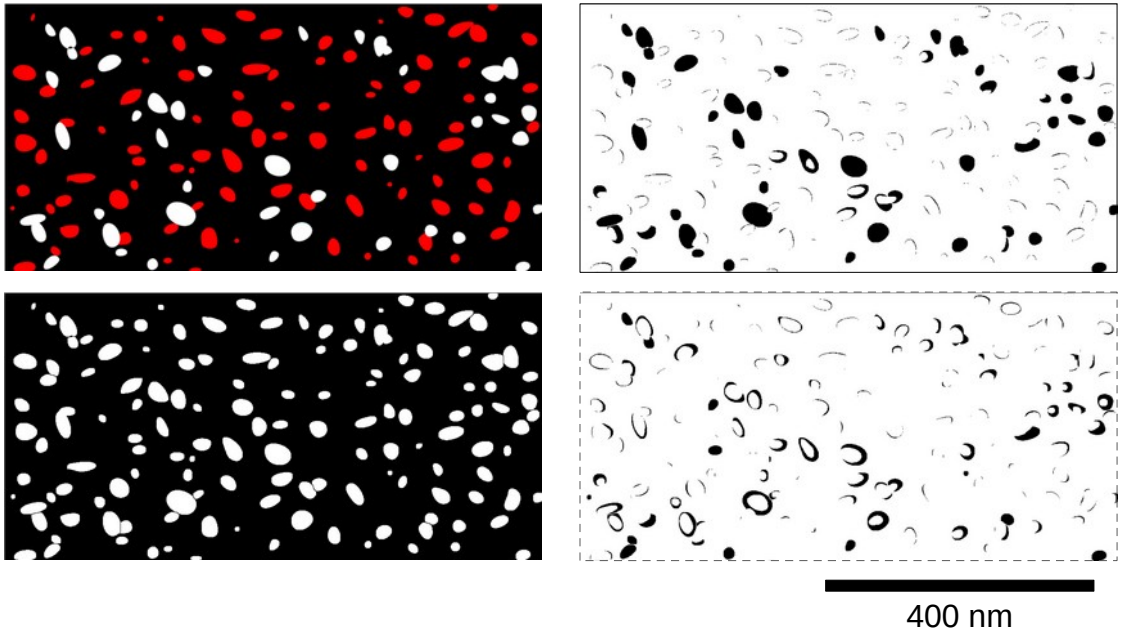






HT analysis of loop dynamics in irradiated MPEAs using DL
(Fine-tuning 1)

Novel Ni-based alloys, 2 MeV Ni^{2+}



Base model (March 2025)
Dataset: 43835 + 10975 loops

Detects 452 loops; (CI > 0.8)

Dataset: 7096 + 1884 loops

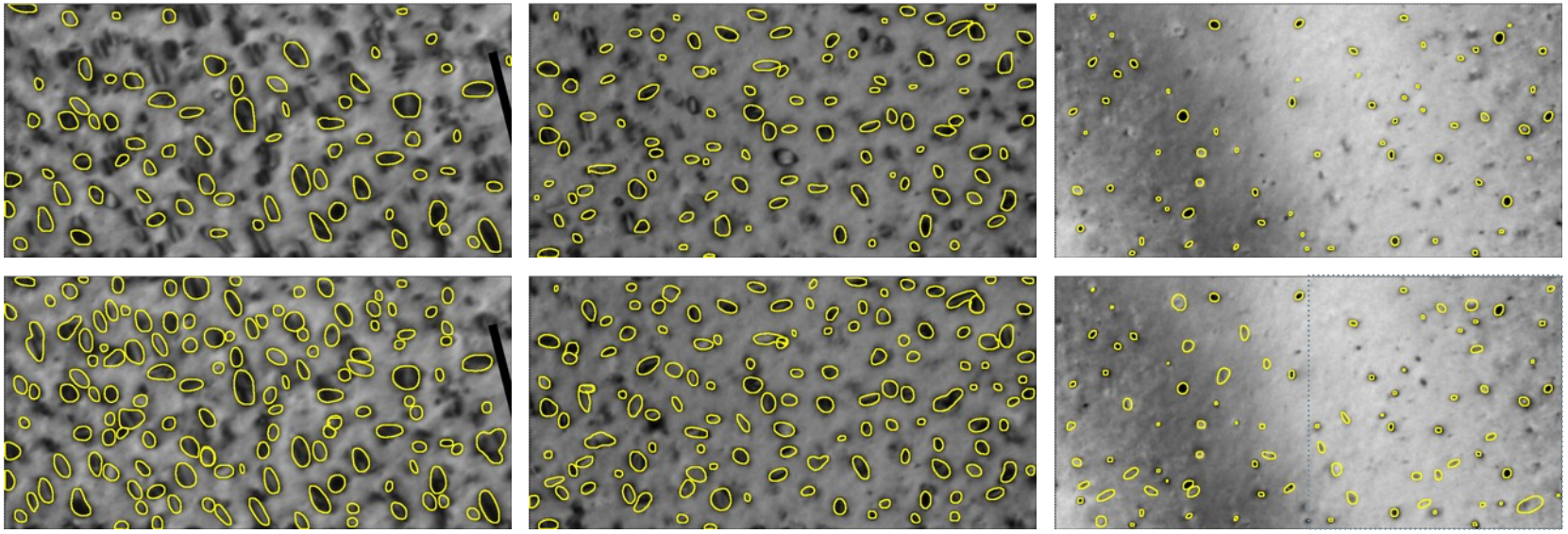
Detects 635 loops; (CI > 0.8)

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HT analysis of loop dynamics in irradiated MPEAs using DL
(Fine-tuning 2)

Novel Ni-based alloys, 2 MeV Ni^{2+}

Before fine-tuning



After fine-tuning

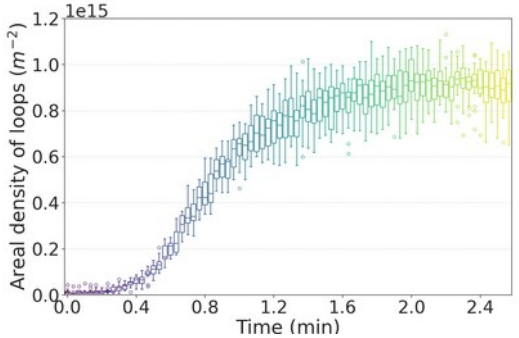
HT analysis of loop dynamics in irradiated MPEAs using DL
(Conclusions)

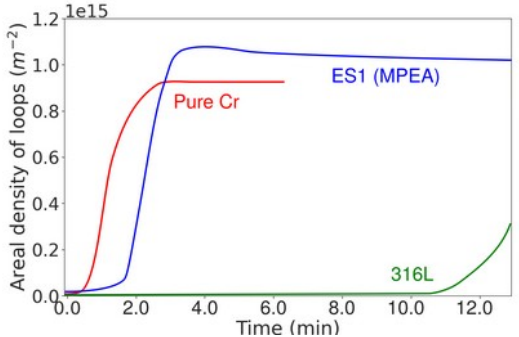
Backbone for Mask R-CNN:
- ResNeXt-101 (best), ResNet-50 (acceptable);
- High-quality segmentation (sizes and the segmented area);
- No need of background subtraction, SVD, etc;

When comparing different materials:
- Employ a common model, threshold (> 0.7) and FOV;

Processing time (on consumer-grade hardware):
- 180 ms per 512x512 px crop (ResNeXt-101);
- 120 ms with ResNet-50 (2s for a 2048 x 2048 image);

Real-time (on-the-fly) segmentation:
- Compromise N of crops to increase the N frames processed.





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HT analysis of loop dynamics in irradiated MPEAs using DL
(Conclusions)

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Real-time (on-the-fly) segmentation:
- Compromise N of crops to increase the N frames processed.

This work was submitted to Micron (Elsevier)
The codebase can be found at:
https://github.com/ai-atoms/creme

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Thank you!

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FEMS 2025 EUROMAT

HT analysis of loop dynamics in irradiated MPEAs using DL
(Mask R-CNN, simplified)

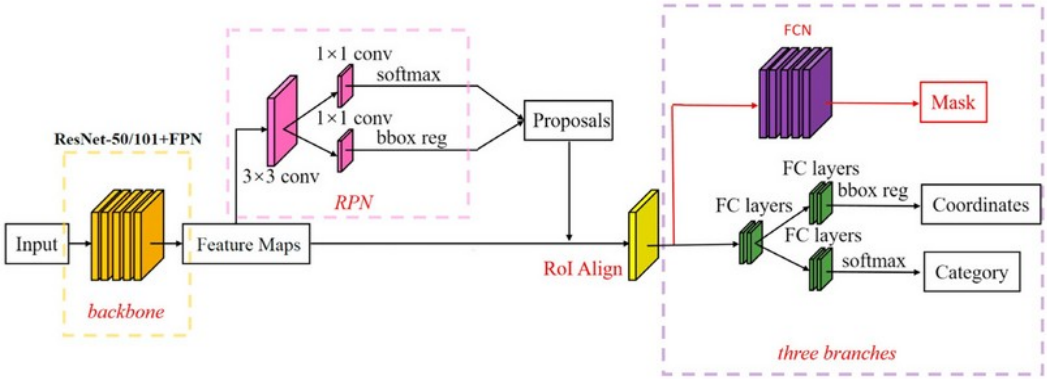


Image source: Lin et al., Forests 13(10), 1603, (2022)

FPN (P2 – P5) and the Region Proposal Network scales:
- 0.250 (1/4), 0.125 (1/8), 0.0625 (1/16), 0.03125 (1/32);
- i.e. ROI pooler handles features in different scales of the pyramid.

Mask R-CNN, x101, 32x8d, FPN3x

Architecture:

- Backbone ResNeXt with 101 layers;
- 32 groups, bottleneck width = 8;
- FPN: Feature Pyramid Network
  - 3x3 conv;

- ROI Pooler + ROI Align
  - 7x7x256 for classification;
  - 14x14x256 for masks;

Total ~107 M (backbone = 90M).

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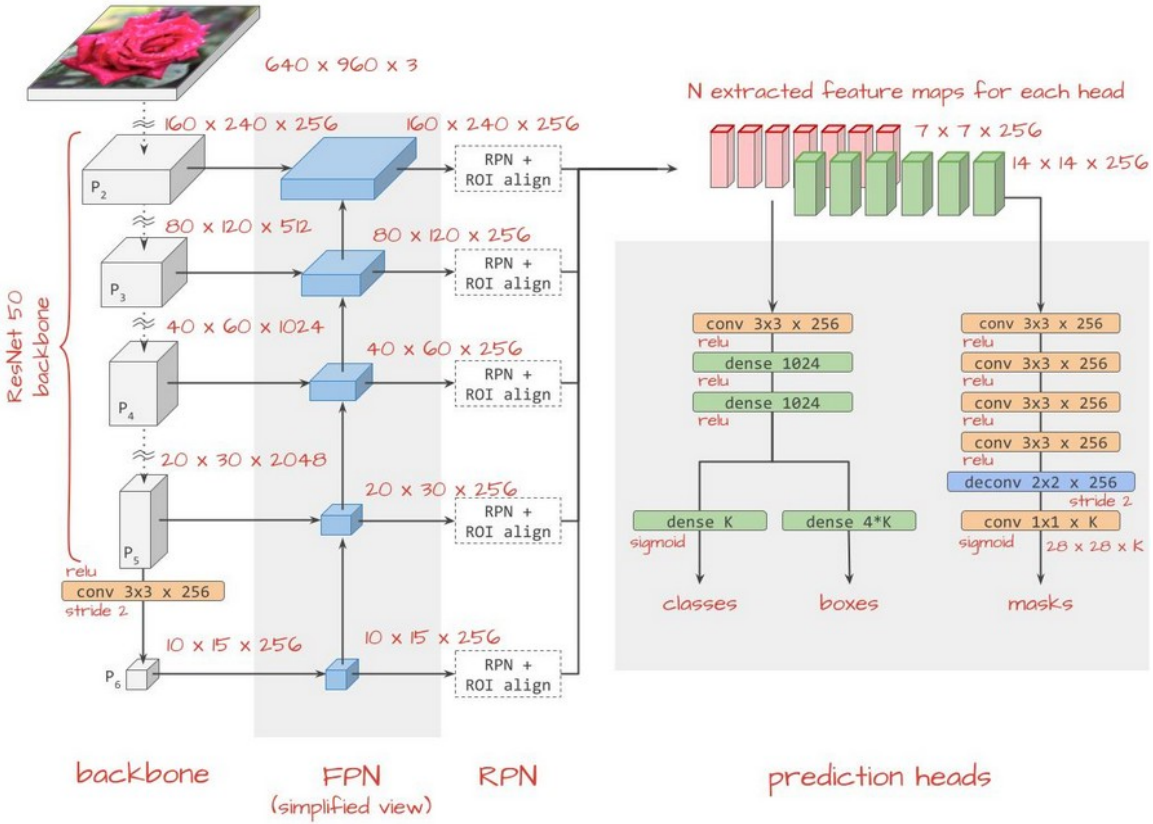


Image source:
x.com/martin_gorner (2021)